1. Fourier Series

If someone gives us a curve, it turns out that a pretty fundamental thing we might want to do is express that curve as the sum of sines and cosines. It's fundamental to quantum mechanics, to signal processing, and, not surprisingly, to the physics of music. For instance, let's say someone plucks a string. How would we expect that to sounds, and how would the sound quality depend on *where* we plucked the string? We'll see that being able to describe a curve a a sum of sines/cosines is key to answering those questions quantitatively. The general class of techinques is called Fourier analysis, after the French mathematician who developed tools to describe how heat flows inside an object.

We've seen that a string undergoing a sinusoidal oscillation produces a pure tone. Further, a string can have many sinusoidal oscillations going on at the same time with different wavelengths/frequencies (the harmonics), and it will produce a sum of pure tones. The ratio of the amplitudes of the harmonics is what sets the timbre of the string. If I pull a string up and let it go, which is what we do when we pluck it, if I knew the amplitude of each harmonic then I can tell you what that plucked string will sound like. If I knew how to describe the shape of the string at the moment you let go as a sum of sines/cosines, then we're done. A complete treatment of Fourier analysis requires some pretty advanced math and is way, way beyond the scope of this course. However, we can do everything we need using just the algebra/trig identities we've already seen (and a bit of cleverness).

2. Sine Series

We'll start with the case of a string clamped to zero at the ends (which is true for musical instruments), and we'll start by considering only waves that have an integer number of wavelengths on our string. You might rightfully object that we can have half-wavelengths as well, but don't worry, that will turn out to be an easy fix.

We could use either sines or cosines (or complex exponentials!) to describe our string, but the edges are a bit simpler if we use sines so we'll do that. Say we have a string of length L and that there is a single sine that has k periods in the string and has amplitude c_k. Then the height of the string $z(x)$ at position x is:

$$z(x) = c_k \sin(2\pi kx/L) \tag{1}$$

When $x = 0$ the sine is zero because $\sin(0) = 0$. When $x = L$ then we have $\sin(2\pi kL/L) = \sin(2\pi k)$. If k is an integer then $z = 0$ at $x = L$, and we have exactly k periods of the sine wave between 0 and L. This will be true for any positive integer value of k. I don't need to worry about $k = 0$ because $\sin(0) = 0$, and I don't need to worry about negative values for k because $\sin(-kx) = -\sin(kx)$ so I can just flip the sign of c_k. We can have any positive integer values of k, though, so the most general thing I can describe is just a sum over values of k:

$$z(x) = \sum_{k=1}^{N} c_k \sin(2\pi kx/L) \tag{2}$$

In words, if you give me a list of the c_k coefficient values, then I start with $k = 1$, multiply the first value c_1 by a sine with one full period in our region. Then I take the second c_2 value, multiply that by a sine with two periods, and add it to the one-period sine I already have. For $k = 3$, I multiply c_3 by the 3-period sine and add it to the one- and two-period sine sum. We keep going until we run out of c_k values, or get bored.

3. Finding the Coefficients

Now that we've decided how we're going to approximate our function, if we knew what the c_k values are, we'd be done. It's not obvious at first glance how to get them for a given function, but there's a clever trick we can play. What happens if we take our function, and multilpy by a sine wave that has say k' full periods in our interval? We take our old function $z(x)$ and multiply by this new sine wave to get $z(x)\sin(2\pi k'x/L)$. We then take the *average value* of this new function. How does this help? Well, let's say our original function had just a single term in it - a sinewave with k full periods. We could pick any value for k we wanted to, but if it helps, you could decide that $k = 3$ and we have three periods of a sine wave. What happens when we multiply that by the sine wave with k' periods? We have

$$\sin(2\pi kx/L)\sin(2\pi k'x/L) \tag{3}$$

We can make this even easier to look at by setting $L = 2\pi$. Nothing changes conceptually if we have a different value, but the math cleans up when we make this choice. We now have

$$\sin(kx)\sin(k'x) \tag{4}$$

Now we're going to use trig identities to work out what the average value of this is given k and k'. Remember that $\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b)$. Similarly, $\cos(a-b) = \cos(a)\cos(b) + \sin(a)\sin(b)$. If I subtract these two then I get:

$$\cos(a-b) - \cos(a+b) = \cos(a)\cos(b) + \sin(a)\sin(b) - (\cos(a)\cos(b) - \sin(a)\sin(b)) \tag{5}$$

$$= 2\sin(a)\sin(b) \tag{6}$$

If I divide by 2 I get the identity that

$$\sin(a)\sin(b) = \frac{1}{2}\left(\cos(a-b) - \cos(a+b)\right) \tag{7}$$

If we take this identity and apply it to Equation 8 then we have

$$\sin(kx)\sin(k'x) = \frac{1}{2}\left(\cos((k-k')x) - \cos((k+k')x)\right) \tag{8}$$

Since k and k' are both integers, we're now taking an average over an integer number of wavelengths of cosine, and the average value of cosine is zero. The only exception is if we're taking the cosine of $0 * x$, because $\cos(0) = 1$, so the average of that is going to be 1. We said that k and k' are both positive, so $k + k'$ is always positive, but $k - k'$ can be zero if $k = k'$. That leaves us with the average of $\sin(kx)z(x) = \frac{c_k}{2}$. We can find all of our coefficients just by averaging our function times a sine.

4. Half-Wavelengths

We've assumed that we have an integer number of wavelengths on our string, but of course a string tied at both ends can also have half-wavelength oscillations as well. The same trick with the same math works if we make k and k' both half-integers in Equation 8. In that case, $k - k'$ and $k + k'$ are both integers (*e.g.* $\frac{3}{2} + \frac{1}{2} = 2$ and $\frac{3}{2} - \frac{1}{2} = 1$). We again average to zero unless $k = k'$.

The final case to worry about is if k is an integer, and k' is a half-integer. In that case, $\sin(kx)$ is what is called an *odd function*. If I take a function and center on some point (in this case, the half-way point of the string), then a function is odd if the right side of the function is a negative copy of the left side, flipped left-right. If I have a single period, for instance, the left half is a positive half-wave, and the right have is a negative half-wave. Since the two halves are equal and opposite, the average has to be zero. A function is *even* if the right half is the left half flipped left-right, but not flipped up-down. The average of an even function is not in general zero. A wave with half-periods on the string will be even because if I flip *e.g.* a half-period left-right, I end up with the same thing. Unlike with numbers, an odd function times an even function is an odd function, and so the average will be zero.

When we try to average $\sin(kx)\sin(k'x)$ but k is an integer and k' is a half-integer, then we're averaging an odd times an even function, and that has to go to zero. So in fact the same averaging trick works and we can find all the coefficients of our function $z(x)$ by averaging $z(x)\sin(kx)$ where k is either an integer or half-integer.

5. Evolution of a Plucked String

We now have the tools we need to understand how a string will behave. We'll just consider the case where we pluck a string, so it starts with some non-zero position (except at the ends!) and zero velocity. The different sine waves behave independently, so if we can write our string height (and putting the string length back in) as

$$\sum c_k \sin(\pi k x/L) \tag{9}$$

then we move each term forward independently in time. We know the modes of a string are $\sin(kx)\sin(\omega t)$ where $\omega/k = v$ or $\omega = kv$ where v is the wave speed on the string. We use Equation 9 to find

$$\omega = \pi k v/L \tag{10}$$

where k is an arbitrary integer. That lets us find what the string looks like at all times:

$$z(x,t) = \sum c_k \sin(\pi k x/L)\cos(\pi k v t/L) \tag{11}$$

If what we care about is music, we want to know what frequencies are produced. Equation 11 tells us just that. The frequencies $\nu = \omega/2\pi$ are given by $\pi k v/L/2\pi = kv/2L$, as we have seen before. The relative amplitude of each frequency is just given by the different c_k. How that converts to sound will depend on the instrument the string is connected to, but the c_k do tell you the total

amount of energy that could come out at each frequency. In particular, if we pluck at different places along the string, the relative c_k will change, and we can say how the sound quality will change.

6. Hammered String

We have so far assumed we have a string starting from a non-zero position with zero velocity. It's pretty common that we start with zero position but non-zero velocity, though. Picture a piano hammer - when it hits the string, the string where the hammer hits picks up a velocity. The string hasn't had time to move, though, so the starting conditions are motion and not position. We do more-or-less the same thing, but now our time part is $\sin(\omega t)$ instead of $\cos(\omega t)$. At $t = 0$, the sin is zero but has non-zero slope in time, or velocity. At $t = 0$, the cosine is non-zero, but it's slope in time is zero.

Out problem is as before, where we want to find the c_k that give

$$\sum c_k \sin(\pi k x/L) \sin(\omega t) \tag{12}$$

gives us the initial velocity. It turns out that the velocity from $\sin(\omega t)$ is just $\omega \cos(\omega t)$ (if you know calculus, this is just the derivative), so we're left with finding

$$v(x) = \sum c_k \sin(\pi k x/L) \cos(\omega t)\omega \tag{13}$$

What Equation **??** tells us is that we do almost exactly the same thing. We start with the velocity as a function of position instead of the height, then we find the average values as usual. The only wrinkle is that the average gives us not c_k but ωc_k, so we have to divide by the ω for that mode.

For a fully general case where you start with non-zero position and non-zero velocity, you just find the two pieces separately. The non-zero position with zero velocity gives you one set of c_k's to match the position, the zero position with non-zero velocity gives you a *different* set of c_k's, and the total solution is the sum of the two pieces. If you do this, don't forget that the time part of the position term has $\cos(\omega t)$ and that the velocity part has $\sin(\omega t)$ in it!

7. Discussion

We've seen that we can write functions as a sum of sines/cosines. For the case of a string tied at both ends, we only get sines, and we saw how to work out the amplitude of each sine wave. We would use different combinations for different setups (*e.g.* a pipe open at both ends would give us cosines), but the math to work out the coefficients is the same. Expressing a function this way is called a *Fourier series* after the great French mathematician (who also came up with dimensional analysis).

There are some general properties it's worth mentioning. If we only use long-wavelength sine waves, we can't make sharp bends. The sum of smooth functions is smooth. If our starting position for a string has a kink in it (say because we plucked with a narrow pick), then we'll need to use a lot

of different sines to match that kink. In turn, our instrument will be producing higher harmonics and give us a brighter sound. Where we pluck makes a difference, too. If we pluck in the center, then the starting position is an even triangle. That means all the odd functions will average to zero, so we'll only get half-integer harmonics. That will absolutely change the sound quality. In general, as you pluck closer to the end, you'll increase the number of higher harmonics as well, giving a brighter sound.

If you wanted to calculate the c_k *really* fast, you could use what's called a Fourier transform. It's a little tricky on a string with half-wavelengths, but the results would be more-or-less instant. If you want to do this, come talk to me. It's less obvious, but the same averaging trick will work on *any* configuration - including 2 and 3-dimensional objects. For those who have taken linear algebra/PDEs, we can write a discretized version of the acceleration operator as a matrix, and the corresponding modes are the eigenvalue/eigenvectors of that matrix, in much the same way as you would solve a system in quantum mechanics. Because eigenvectors are always orthogonal, once we have them, we can always find the coefficients by averaging our function times the corresponding mode. The (square roots of the) eigenvalues tell you the frequencies at which those modes will vibrate, which in turn lets you predict your instrument's sound. To those for whom the previous sentence looked like gibberish, this is why I said to take on faith the modes of more complicated setups when I give them to you!